
Abstract

This report describes a molecular dynamics simulation of Argon atoms. We used a Lennard-Jones potential and the velocity-Verlet algorithm for the equations of motion. All calculations were done in reduced units that are practical and intuitive for the system at hand. To reproduce the infinite nature of the system, periodic boundary conditions in combination with the minimal image convention were used. Three state points were calculated, each corresponding to the three different phases: gas, liquid and solid. These phases could be clearly identified by their pair correlation function. They aligned with the expected behaviour from literature. The calculated pressure values for the gas and liquid states align with the values found in literature for a Lennard-Jones fluid, which validates our simulation. Total energy conservation was also obtained after equilibrium. Identifying the triple-point was attempted, but We have too little sampled points to identify a clear phase boundary. Our phase identification method is also too simplistic. Overall, the simulations returned the expected behaviour of a system of Argon atoms with a Lennard-Jones potential.

Contents

1	Introduction	3
2	Theory	3
2.1	Newton's equations for many interacting particles	3
2.2	Lennard-Jones potential and derived force	4
2.3	Periodic boundaries and minimum image convention	4
2.4	Velocity Verlet integration and timestep choice	5
2.5	Reduced units	6
2.6	Observables: energy, pair correlation function, and virial pressure .	7
3	Methods	8
3.1	Initial conditions	9
3.2	System simulation	10
3.3	Calculating observables	11
3.4	Using $g(r)$ and pressure to distinguish phases	11
4	Results	12
4.1	Pressure	12
4.2	Pair correlation function	13

4.3	Phase diagram of argon	13
4.4	Total energy of the system	14
5	Discussion	15
5.1	Pressure values	15
5.2	Pair correlation function	15
5.3	The triple point of argon	15
5.4	Energy conservation	15
6	Summary	16
	References	16

1 Introduction

Molecular dynamics can be used as a method to simulate a molecule or atom, where solving a system analytically would be too complex. It is a computational technique that relies on numerically solving for the motion of the particles in the system, assuming Newtonian physics. This then allows for particle-potential interactions. We will simulate Argon in different phases, based on the provided pressure and density.

The atoms will be represented as particles, which will interact with a potential. The potential used in the simulation is the Lennard-Jones potential, which is explained in detail later. In this case Argon is chosen since it is a noble gas, so it does not form molecules and is studied extensively. Also Helium would have additional dipole interactions. To simulate an infinite system we make use of periodic boundaries. To mimic interactions, the minimal image convention is used.

The units of the variables we encounter in this simulation are not necessarily intuitive for the problem. Therefore we use reduced units that are easier to work with and allow for more insightful interpretations of the results. The energy conservation is imposed by using the velocity-Verlet algorithm to calculate the equations of motion.

In the following sections we will study the properties of a system of Argon atoms with a Lennard-Jones potential. Then we will calculate the pair correlation function and pressure for three different state points, corresponding to the three different phases: gas, liquid and solid.

2 Theory

2.1 Newton's equations for many interacting particles

From Newtonian particle dynamics with potential $U(x)$,

$$\mathbf{F}(x_i) = m\mathbf{a} = m\frac{d^2x_i}{dt^2} = -\nabla U(x_i). \quad (1)$$

For more particles we have to look at the total potential,

$$U_{\text{tot}} = \sum_{i < j} U(|x_i - x_j|), \quad (2)$$

so

$$\mathbf{F}(x_i) = -\sum_j \nabla U(|x_i - x_j|). \quad (3)$$

2.2 Lennard-Jones potential and derived force

In this simulation we use the Lennard-Jones potential

$$U(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]. \quad (4)$$

This potential models both the van der Waals interaction and Pauli repulsion.

2.3 Periodic boundaries and minimum image convention

As for the boundary conditions of the system, we introduce periodic boundaries, as to simulate an infinite system (no wall-interactions). We then use the minimal image convention. One can imagine this as follows. When we simulate a box with periodic boundary conditions, any particle that leaves the box will enter it on the opposite side. This means we have 'infinitely' many images of the system. The easiest way to imagine this is a bead moving around a 1D chain. Defining a starting point, if the bead moves to the end of the chain, imposing periodic boundary conditions simply corresponds to wrapping the chain in a loop. The velocity then remains unchanged.

This is necessary because without this there would be less particles to interact with around the edge compared to particles in the center of the box, which would be unphysical (since the box is just a computational limitation, not a physical wall).

The downside of this is that we now have to consider all these particles (that should be there). Luckily our potential, the Lennard-Jones potential, decreases by r^{-6} . So as long as we choose a box size that is reasonably bigger than the interaction length we can safely only consider the nearest by image particle. This still increases the quality of our simulation. So two particles close to the left and right boundary now interact with the image particle that is way closer to them than the separation of the original two particles. To determine this

Mathematically this boils down to minimizing over the neighbouring periodic images.

$$r_{ij} = \min_{k,\ell,m=-1,0,1} \sqrt{(x_i - x_j + kL)^2 + (y_i - y_j + \ell L)^2 + (z_i - z_j + mL)^2} \quad (5)$$

Since the box is orthogonal this reduces to checking three different 1D cases.

$$x_{\text{closest}} = x_j + \begin{cases} 0 & \text{if } -\frac{L}{2} < x_i - x_j < \frac{L}{2} \\ L & \text{if } x_i - x_j > \frac{L}{2} \\ -L & \text{if } x_i - x_j < -\frac{L}{2} \end{cases} \quad (6)$$

One can see that this either picks the closest original particle or the nearest image ($\pm L$).

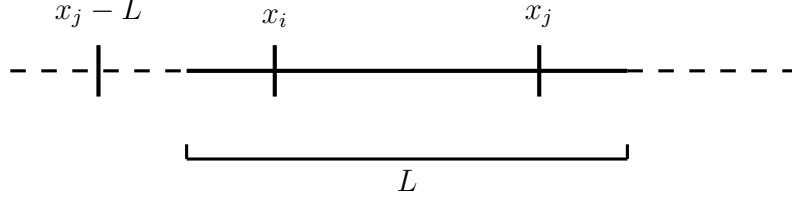


Figure 1: An example of the case where $x_i - x_j < -L/2$. As you can see the closest particle is at $x_j - L$, not x_j .

2.4 Velocity Verlet integration and timestep choice

To solve for the equations of motion, we can Taylor expand the position. Using Verlet's algorithm, we can do this by expanding both forward and backward positions:

$$x(t+h) = x(t) + h\dot{x}(t) + \frac{h^2}{2}\ddot{x}(t) + \frac{h^3}{3!}\frac{d^3}{dt^3}x(t) + \mathcal{O}(h^4), \quad (7)$$

$$x(t-h) = x(t) - h\dot{x}(t) + \frac{h^2}{2}\ddot{x}(t) - \frac{h^3}{3!}\frac{d^3}{dt^3}x(t) + \mathcal{O}(h^4). \quad (8)$$

Adding these gives

$$x(t+h) + x(t-h) = 2x(t) + h^2\ddot{x}(t) + \mathcal{O}(h^4), \quad (9)$$

so

$$x(t+h) = 2x(t) - x(t-h) + \frac{h^2}{m}F(x(t)) + \mathcal{O}(h^4). \quad (10)$$

And

$$v(t) = \frac{x(t+h) - x(t-h)}{2h} + \mathcal{O}(h^2), \quad (11)$$

which is naturally symmetric in time. By Noether's theorem, this ensures conservation of energy.

We use the form with $v(t)$ explicitly,

$$x(t+h) = x(t) + hv(t) + \frac{h^2}{2m}F(x(t)) + \mathcal{O}(h^3), \quad (12)$$

and

$$v(t+h) = v(t) + h\dot{v}(t) + \frac{h^2}{2}\ddot{v}(t) + \mathcal{O}(h^3). \quad (13)$$

We want to express this in terms of the current velocity $v(t)$ and force. So,

$$\dot{v}(t+h) = \dot{v}(t) + \mathcal{O}(h^2), \quad (14)$$

and

$$\ddot{v}(t+h) = \frac{1}{h} \frac{1}{m} [F(x(t+h)) - F(x(t))]. \quad (15)$$

Finally,

$$v(t+h) = v(t) + h\dot{v}(t+h) - h\dot{v}(t) + \mathcal{O}(h^2) + \frac{h^2}{2}\ddot{v}(t) + \mathcal{O}(h^3) \quad (16)$$

$$= v(t) + \frac{h}{m}F(x(t+h)) - \frac{h^2}{2}\ddot{v}(t) + \mathcal{O}(h^3) \quad (17)$$

$$= v(t) + \frac{h}{m}F(x(t+h)) - \frac{h}{2m} [F(x(t+h)) - F(x(t))] + \mathcal{O}(h^3) \quad (18)$$

$$= v(t) + \frac{h}{2m} [F(x(t+h)) + F(x(t))] + \mathcal{O}(h^3). \quad (19)$$

A sensible choice of size for the time-step h is argued in the following section.

2.5 Reduced units

For the physics we are simulating, the standard SI-units are not really intuitive since they vary by orders of magnitude. We therefore introduce reduced units that correspond to characteristic scales of the system:

$$\tilde{x} = \frac{x}{\sigma}, \quad \tilde{r} = \frac{r}{\sigma}, \quad \tilde{U}(\tilde{r}) = \frac{U(r)}{\epsilon} = 4 \left(\frac{1}{\tilde{r}^{12}} - \frac{1}{\tilde{r}^6} \right). \quad (20)$$

Then

$$\frac{d^2x}{dt^2} = \frac{1}{\sigma} \frac{d^2\tilde{x}}{dt^2} = -\frac{1}{\sigma m} \nabla U(r) = -\frac{\epsilon}{\sigma m} \nabla \tilde{U}(\tilde{r}) = -\frac{\epsilon}{\sigma^2 m} \tilde{\nabla} \tilde{U}(\tilde{r}), \quad (21)$$

so time transforms like

$$\tilde{t} = t \sqrt{\frac{\epsilon}{\sigma^2 m}}. \quad (22)$$

This results in

$$\frac{d^2\tilde{x}}{d\tilde{t}^2} = -\tilde{\nabla} \tilde{U}(\tilde{r}), \quad (23)$$

and

$$\tilde{v} = \frac{d\tilde{x}}{d\tilde{t}} = \frac{dx}{dt} \sqrt{\frac{m}{\epsilon}}. \quad (24)$$

Which we can also express, using the equipartition theorem like:

$$\tilde{v} = \sqrt{\frac{3k_b T}{\epsilon}}. \quad (25)$$

This is exactly the same equation as before, with the understanding that we now measure distance in units of σ , energies in units of ϵ , and time in units of $\sqrt{m\sigma^2/\epsilon}$.

This also allows us to take a sensible time step length, e.g. 1/100 in our reduced units.

We can then continue with all our expressions, keeping the above in mind. So e.g. the factor $1/m$ disappears and all variables get a tilde.

2.6 Observables: energy, pair correlation function, and virial pressure

The density of the system can be set. Given the amount of atoms and the mass of each atom, simply use

$$\rho = \frac{m}{V} = \frac{N m_A}{L^3}, \quad (26)$$

so

$$\tilde{\rho} = \frac{\rho \sigma^3}{m} = \frac{N}{\tilde{L}^3}. \quad (27)$$

The simulation uses a lattice of 3^3 unit cell boxes (4 atoms each). Thus we simulate 108 atoms.

We draw velocities from the Maxwell–Boltzmann distribution,

$$p(v_x) \propto e^{-\frac{mv_x^2}{2k_B T}}, \quad (28)$$

and in reduced units the exponent becomes

$$\frac{mv_x^2}{2k_B T} = \frac{\tilde{v}_x^2}{2\tilde{T}}. \quad (29)$$

To push this towards equilibrium, we rescale the velocities in the beginning of the simulation. We expect

$$E_{\text{kin}} = \frac{3}{2}(N-1)k_B T \quad (30)$$

by the equipartition theorem.

Now rescale $v_i \rightarrow \lambda v_i$ for the ratio

$$\lambda = \sqrt{\frac{E_{\text{target}}}{E_{\text{current}}}} = \sqrt{\frac{3k_B(N-1)T}{\sum_i m v_i^2}}. \quad (31)$$

And in our reduced units ($k_B = 1$, $m = 1$) we get

$$\lambda = \sqrt{\frac{\frac{3}{2}(N-1)\tilde{T}}{\tilde{E}_{\text{kin}}}}. \quad (32)$$

Using a pair correlation function we can measure the level in which the particles are organised. This is done with the help of a histogram $n(r)$ which counts the distances between all pairs of particles. For a solid (crystal) we then expect $n(r)$ to be clearly peaked. For a gas we expect it to be spread and weak. We can then take the average of $n(r)$ over multiple configurations, and normalise it to 1 for a homogeneous distribution to get

$$g(r) = \frac{2V}{N(N-1)} \frac{\langle n(r) \rangle}{4\pi r^2 \Delta r}. \quad (33)$$

The pressure can be determined with the following formula given in the lectures:

$$\frac{mP}{k_B T \rho} = 1 - \frac{1}{3Nk_B T} \left\langle \frac{1}{2} \sum_i \sum_{j>i} r_{ij} \frac{\partial U(r_{ij})}{\partial r} \right\rangle. \quad (34)$$

Or in reduced units:

$$\tilde{P} = \tilde{\rho} \tilde{T} \left(1 - \frac{1}{3N\tilde{T}} \left\langle \frac{1}{2} \sum_i \sum_{j>i} \tilde{r}_{ij} \frac{\partial \tilde{U}(\tilde{r}_{ij})}{\partial \tilde{r}} \right\rangle \right). \quad (35)$$

3 Methods

In order to simulate a system of 108 Argon atoms, we first create semi-randomized initial conditions, then we attempt to equilibrate the system by using the Velocity relaxation strategy. Then we simulate the system for a set amount of time(steps), from which we will draw samples of the system to use in calculating observables.

All physical quantities are their dimensionless version. All distances between particles are calculated using the minimum image convention. All forces and potential energies are derived from the dimensionless Lennard-Jones potential.

We will simulate the system in three different states: Gas: $\rho = 0.3$, $T = 3.0$. Liquid: $\rho = 0.8$, $T = 1.0$. Solid: $\rho = 1.2$, $T = 0.5$. Each state is simulated 10 times, over which we average to obtain the results.

For the attempted measurement of the triple point, we do identical simulations, except now we range over large values of ρ and T . We don't repeat any measurements to average over.

3.1 Initial conditions

The initial conditions are completely determined by the density and temperature variables. The density determines the length of the system. The crystal structure of argon is face-centered cubic, which we will also use to define the start positions. One way of positioning each atom nicely is by defining a cell of 4 atoms and placing these cells on a simple cubic lattice. To get a system of 108 particles, we place 3 cells per axis ($4 \cdot 3^3 = 108$). To define the starting velocities, we draw random samples from the Maxwell-Boltzmann distribution.

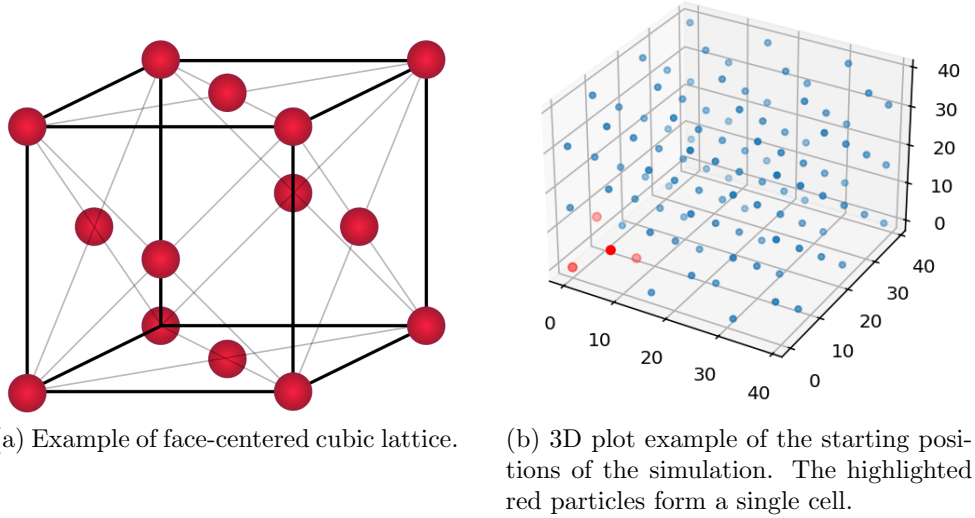


Figure 2: Visual representation of the face-centered cubic lattice. On the left is the general visualization of the FCC lattice. On the right is the implementation of the FCC lattice in the simulation.

3.2 System simulation

We simulate the system for 1500 timesteps of unit 0.01. For the first 500 timesteps we apply velocity relaxation.

Per timestep we do the following procedure. First we save the positions and velocities of all particles. Secondly, we calculate the kinetic and potential energy of the system. The kinetic energy is the sum of all kinetic energies of individual particles ($\frac{1}{2}mv^2$). We then save the kinetic and potential energies. Lastly, we calculate the new positions and velocities. We first calculate the forces on the current positions, then we can calculate the new positions using the current positions, velocities and forces. Using the new positions, we can calculate the forces on the new position, which we can use to calculate the new velocities.

While velocity relaxation is enabled, we rescale the velocities every 100 timesteps. For this we use the above calculated kinetic energy.

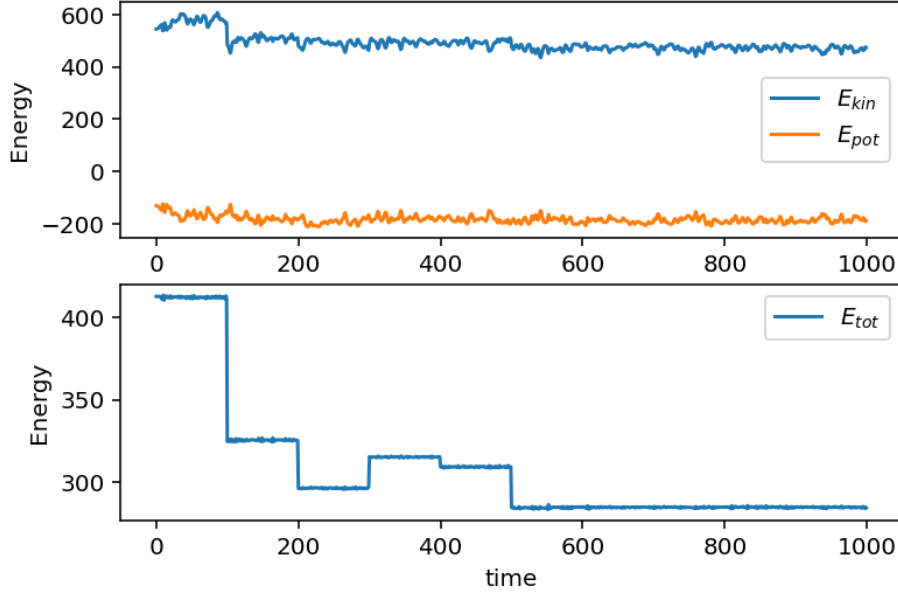


Figure 3: An example of the energy of the system in time. In the first 500 timesteps, velocity relaxation is enabled and occurs every 100 timesteps. We can see how the Kinetic energy shifts, which results in a jump in total energy. The kinetic energy will always fluctuate, thus there will always be a possibility of rescaling the velocities to the desired value. However, we see that only the first jump has a major impact on the system, while the next few are smaller and much closer together. This specific simulation was done in the gas phase. (Note: units are the reduced units as defined in the theory section, so dimensionless)

3.3 Calculating observables

To calculate the pressure and pair correlation function we average over samples taken from the saved data. These samples are taken every 50 timesteps while velocity relaxation is disabled.

3.4 Using $g(r)$ and pressure to distinguish phases

We can use the earlier defined pair correlation function to distinguish the different phases of gas, liquid and solid. This is because each of these phases have characteristic properties that show up on a $g(r)$ plot^[1]. We see that a gas state has a peaked point correlation function but then quickly drops to 1. A liquid state however also shows this initial peak, but then fluctuates around 1, slowly stabilizing to 1 almost like a damped oscillation. Finally, a solid state gives more distinct peaks, because these are the characteristic inter-atomic distances for a solid lattice-like structure. We found an molecular dynamics Argon simulation that also uses the Lennard-Jones potential, which shows exactly the above described behaviour^[3]. We can validate our simulation by checking whether our pair correlation functions for the different states follow these same characteristic properties.

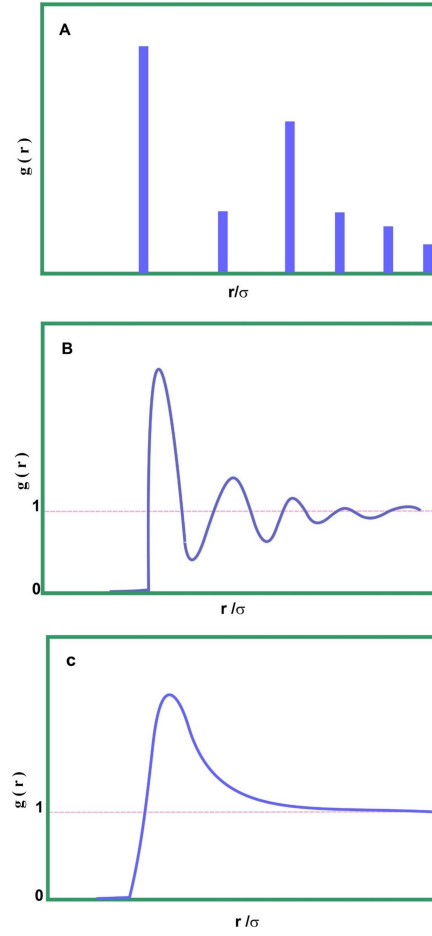


Figure 4: Pair correlation functions of (top to bottom): solid, liquid and gas.^[1]

4 Results

4.1 Pressure

The calculated pressures of the different configurations are as follows:

Phase	$\tilde{\rho}$	$\tilde{T}$	$\tilde{P}$
Gas	0.3	3.0	0.94 ± 0.05
Liquid	0.8	1.0	1.1 ± 0.1
Solid	1.2	0.5	8.6 ± 0.2

Table 1: Reduced pressure for different phases.

4.2 Pair correlation function

The calculated pair correlation functions are shown in figure 5. The errors on the values are the combination of the standard deviation of the averaging of 10 simulations and the propagated standard deviation of one simulation.

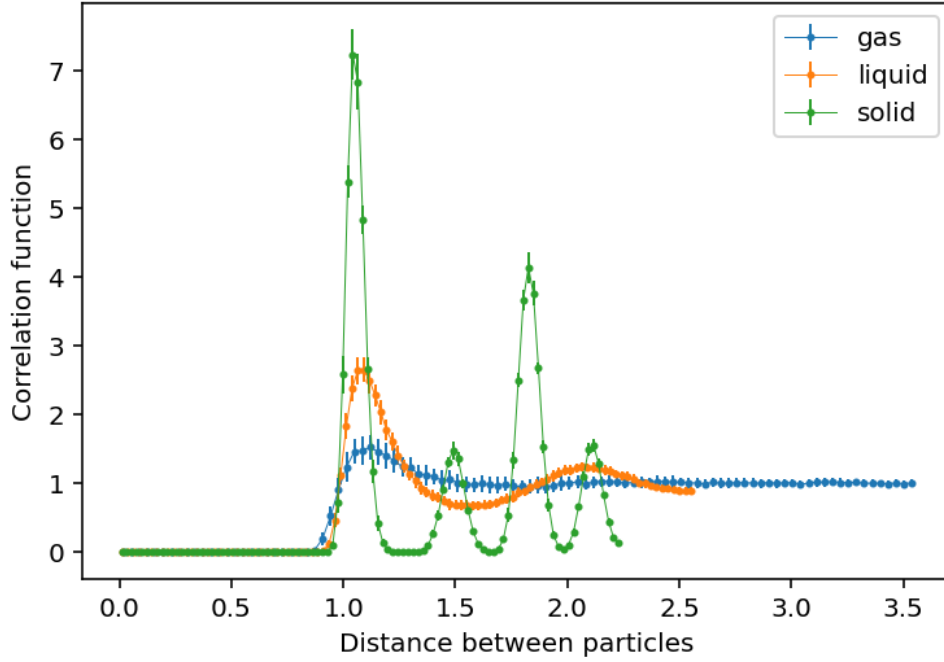
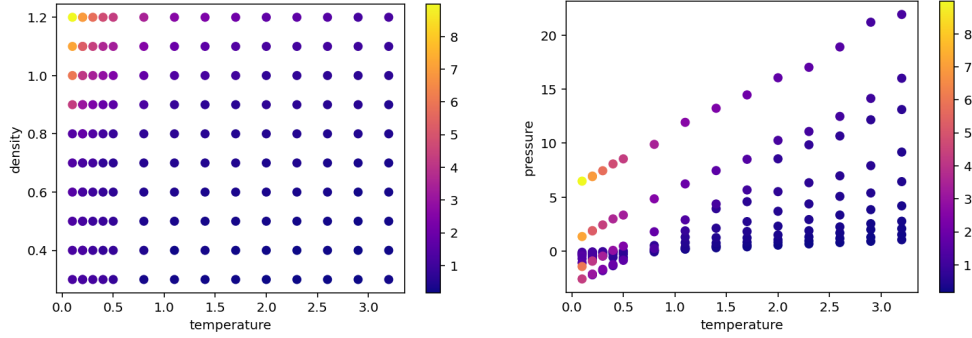


Figure 5: The pair correlation function for the three simulated systems. Here we can clearly the differences between the phases. The gas phase is the most constant, only slightly peaking at the start. The liquid phase has a larger peak, moved more to the right, after which it oscillates. The solid phase has sharp peaks at specific points, while even going to zero in between peaks. The order of magnitude of the errors are ~ 0.2 , with the largest error being 0.4 on the largest peak of the solid. (Note: units are the reduced units as defined in the theory section, so dimensionless)

4.3 Phase diagram of argon

In order to create some sort of phase diagram from the simulation, we opted to plot the difference in maximum and minimum value found in the corresponding correlation function from distances $\tilde{r} > 1.5$. The reasoning behind this is that this value increases as sharper peaks are formed in the pair correlation function, which in turn is caused by a different state of phase. The condition on the distance

causes us to ignore the 'starting' peaks of the pair correlation function, which are present for all phases (only differing in size).



(a) Density-temperature plot of the difference in maximum and minimum value found in the corresponding correlation function from distances $\tilde{r} > 1.5$.

(b) Pressure-temperature plot of the difference in maximum and minimum value found in the corresponding correlation function from distances $\tilde{r} > 1.5$. For low temperatures and low densities, we obtain negative pressure values.

Figure 6: Diagrams plotting the difference in maximum and minimum value found in the corresponding correlation function from distances $\tilde{r} > 1.5$. A high value corresponds to a more solid like pair-correlation function, while a low value corresponds to a more gas like pair-correlation function.

4.4 Total energy of the system

We would also like to add some information about the total energy of the system, which will be relevant in the discussion. The averaged total energy of the different phases are as follows:

Phase	$\tilde{\rho}$	$\tilde{T}$	$\tilde{E}$
Gas	0.3	3.0	479 ± 4
Liquid	0.8	1.0	161 ± 5
Solid	1.2	0.5	80 ± 4

Table 2: Reduced total energy for different phases.

The largest deviations from these averages are 50, 35 and 18 for the gas, liquid and solid phase respectively.

5 Discussion

5.1 Pressure values

When can compare our values for the pressure in the case of $\tilde{\rho} = 0.3$, $\tilde{T} = 3.0$ and $\tilde{\rho} = 0.8$, $\tilde{T} = 1.0$ with the values found in an article that simulates a Lennard-Jones fluid in our reduced units^[2]. They find the following values: $\tilde{P} = 0.999 \pm 0.002$ and $\tilde{P} = 1.011 \pm 0.006$. This is in the same ballpark as our simulated values. In the first case only slightly different and for the latter case actually within the margin of error.

5.2 Pair correlation function

Comparing the above with the shapes of the pair correlation function $g(r)$ found in the literature, we see that our simulation matches these shapes for each phase, strongly suggesting our simulations simulate the different phases properly in a physical manner.

5.3 The triple point of argon

Although the method used to plot a phase-diagram seemed decent at first, it has not amounted to any meaningful result. This is because there are not enough data points to clearly see any phase transition lines and the method to distinguish the phases is also very primitive. We do however see logical ordering of the phases, namely that increasing pressure will push the correlation function to the solid 'side', while increasing temperature will push it to the 'gas' side. We also obtained negative pressure values for some simulations. These do seem consistent in the sense that they follow a linear pattern in the phase diagram. Obviously, this isn't physical, perhaps the formula used to calculate the pressure is not very accurate in the low temperature and density regime.

5.4 Energy conservation

In an ideal physical system, the total energy should be perfectly constant. In our simulation this is not the case. However, we do have energy conservation in the sense that on average the energy is seemingly constant. The fluctuations are also small compared to the total value.

6 Summary

We simulated Argon atoms with molecular dynamics using a Lennard-Jones interaction potential. This captures the Pauli repulsion and Van der Waals attraction. We used periodic boundary conditions and the minimal image convention to account for the finiteness of the simulation size. We only considered nearest neighbour interactions and used a computational trick (orthogonality of coordinates) to simplify the model. To account for conserved quantities, specifically the energy, we used the velocity-Verlet algorithm. Furthermore, we used reduced units that are characteristic and intuitive to the problem. With this model we studied three different phases.

The main result of the simulation is the clear distinction in phases seen in the pair correlation functions. We reproduced the expected behaviour from the literature. The gas phase shows a clear peak and quickly stabilizes to unity, the liquid phase also peaks but then acts like a damped oscillation and finally we get sharp peaks for the solid state. The peaks align with the expected characteristic distances. We also compared our calculated pressure to the pressure found back in literature for a fluid with a Lennard-Jones potential. Our values closely follow the ones found in the literature, which validates the physical validity of the simulation.

As for the energy conservation, the total energy fluctuations were very small compared to the total energy, especially after the equilibrium was reached. The velocity-Verlet algorithm with the chosen timestep thus reproduced a stable system.

Our attempt at identifying the triple-point was not yet successful. We have too little sampled points to identify a clear phase boundary. Our phase identification method is also too simplistic and there is the matter of negative pressure values. Identifying the triple-point will involve improving our simulation and calculation on multiple fronts.

To summarize, we successfully developed a molecular dynamics simulation that aligns with the expected physical properties of an Argon gas in the gas, liquid and solid state.

References

- [1] Elaheh K. Goharshadi. A review on the radial distribution function: Insights into molecular structure, intermolecular interactions, and thermodynamic properties. *Journal of Molecular Liquids*, 433:127900, 2025. ISSN

0167-7322. doi: <https://doi.org/10.1016/j.molliq.2025.127900>. URL <https://www.sciencedirect.com/science/article/pii/S0167732225010773>.

- [2] J. Karl Johnson, John A. Zollweg, and Keith E. Gubbins. The lennard-jones equation of state revisited. *Molecular Physics*, 78(3):591–618, 1993. doi: 10.1080/00268979300100411. URL <https://doi.org/10.1080/00268979300100411>.
- [3] Ethan van Woerkom. The phases of argon. *simulation*, 6:1, 2019.